



蔡詔歲
TSAI, CHAO-WEI

CONTACT

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SKILLS

- Python (base)
- R (base)
- LaTeX (base)
- microsoft SQL (base)
- Image Processing
- Computer Vision



PROFILE

Statistics Master with expertise in ML, Time-Series, and Computer Vision. Proven ability to execute end-to-end projects, including automated satellite imagery pipelines and real-time traffic forecasting. Proficient in Python/R, data imputation, and SHAP analysis, with a focus on converting complex data into actionable insights and interactive dashboards.



RESEARCH PROJECT

2025 Expressway Smart Transportation Management Creative Competition (Innovation Contest - Nominated)

- Web Scraping from Unofficial Sources.
- Processed large-scale heterogeneous datasets from government open data platforms and implemented data imputation strategies
- Designed an interactive dashboard integrating OpenStreetMap to visualize real-time forecasts, providing proactive alerts for congestion events to enhance travel decision-making
- Developed a short-term traffic prediction system using XGBoost and Change Point Detection to identify the onset and dissipation of highway congestion within a 30-minute horizon
- Utilized SHAP analysis to interpret model predictions, identifying that real-time speed drives short-term outcomes, while temporal factors and weather conditions significantly impact mid-to-long-term traffic patterns

Cross-disciplinary Research: Impact of Coronal Hole Evolution on Solar Wind Speed Prediction

- Developed an automated image processing pipeline using the OTSU algorithm to segment solar satellite imagery and quantify coronal hole energy metrics.
- Implemented Sliding Correlation and MSE minimization to calibrate phase shifts between solar observations and solar wind speed, accounting for propagation latency.
- Proposed a customized PILE correction model integrating physical constants and background wind speed to enhance arrival time forecasting accuracy.
- Applied Hampel Filters for outlier detection and conducted window size sensitivity analysis to ensure the stability and generalizability of the forecasting model.



EDUCATION

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| National Changhua University of Education
Department of Mathematics | 2019 - 2023 |
| National Taipei University
Institute of Statistics | 2024 - 2026 |